# Event Density and Core Interaction Laws:

**Extraction and Geometry of the Individuation Manifold in Event Density**

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## Abstract

Event Density (ED) is an architectural field theory in which persistent entities arise from the intrinsic geometry of the update rule. While ED-Arch-01 through ED-Arch-06 established the foundations of individuation, suppression, mobility, and the architectural action, the laws governing interactions between entities remained uncharacterized. Here we present the first explicit mapping of the core-interaction manifold in ED: a smooth, low-dimensional surface in (α, m, d) parameter space that separates the three fundamental interaction regimes—ANNIHILATE, MERGE, and HOVER/ORBIT.

Using a full sweep of drain strength α, mobility exponent m, and core separation d, we extract boundary points where regime transitions occur, revealing a structured topology that includes diagonal stability boundaries, non-monotonic separation windows, and architectural triple-points. We fit a bilinear surface in log-alpha space that approximates the ANNIHILATE boundary, demonstrating that the manifold is geometrically coherent and governed by interpretable architectural flows. The resulting surface provides the first quantitative interaction law in ED, linking curvature accumulation, mobility-weighted freezing, and drain-driven dissipation.

ED-Arch-07 therefore marks a transition in the series: from establishing the architectural foundations of ED to deriving explicit laws of entity–entity interaction. The individuation manifold uncovered here forms the basis for future work on quantization, 3D structure, noise robustness, and the architectural action.

## 1. Introduction

Event Density was introduced as an architectural field theory: a framework in which structure, behavior, and identity arise not from imposed rules but from the intrinsic geometry of the update law. ED-Arch-01 through ED-Arch-06 established the foundations of this program. They defined the architectural primitives (saddles, gradients, boundaries), the regime taxonomy (coherent, marginal, incoherent), the physics layer (suppression, drain, mobility), and the action-based interpretation of ED as a field theory.

Across these earlier works, a recurring theme emerged: ED supports persistent, spatially localized entities—cores—whose existence depends on the balance between curvature-driven accumulation and drain-driven dissipation. Yet while the conditions for individuation were explored, the laws governing interactions between entities remained uncharacterized. How do two cores behave when placed in proximity? Under what conditions do they merge, hover, orbit, or dissolve? And is there a geometric structure that governs these transitions?

This paper answers those questions.
We present the first explicit mapping of the core-interaction manifold in ED: a smooth, low-dimensional surface in (α, m, d) space that separates the three fundamental interaction regimes—ANNIHILATE, MERGE, and HOVER/ORBIT. We extract this manifold directly from simulation data, fit an interpretable architectural surface, and interpret its geometry in terms of the underlying ED flows.

This paper therefore marks a transition in the series: from establishing the architectural foundations of ED to deriving interaction laws—the architectural analogues of forces, potentials, and stability boundaries in physical field theories.

## 2. Background

### 2.1 Architectural Entities in ED

In ED, a "core" is not a particle but a curvature well: a region where density gradients and mobility collapse to form a persistent, self-maintaining structure. ED-Arch-02 formalized the distinction between:

- Coherent regimes, where curvature wells persist
- Marginal regimes, where identity is fragile
- Incoherent regimes, where drain overwhelms structure

Cores are architectural entities: they arise from the rule itself, not from imposed ontology. Their stability depends on the interplay of suppression, mobility, and drain.

### 2.2 Parameters Governing Identity

Three parameters determine whether two cores persist and how they interact:

- α (drain strength): controls dissipation; high α dissolves structure.
- m (mobility exponent): controls freezing; high m suppresses diffusion near saturation.
- d (separation): controls interaction geometry; determines whether curvature wells overlap, compete, or remain distinct.

These parameters define a three-dimensional space of possible interactions. Prior work explored α and m independently, but their joint effect—and the role of separation—had not been mapped.

**Figure 01**: Two-core initial condition, Gaussian wells, separation d

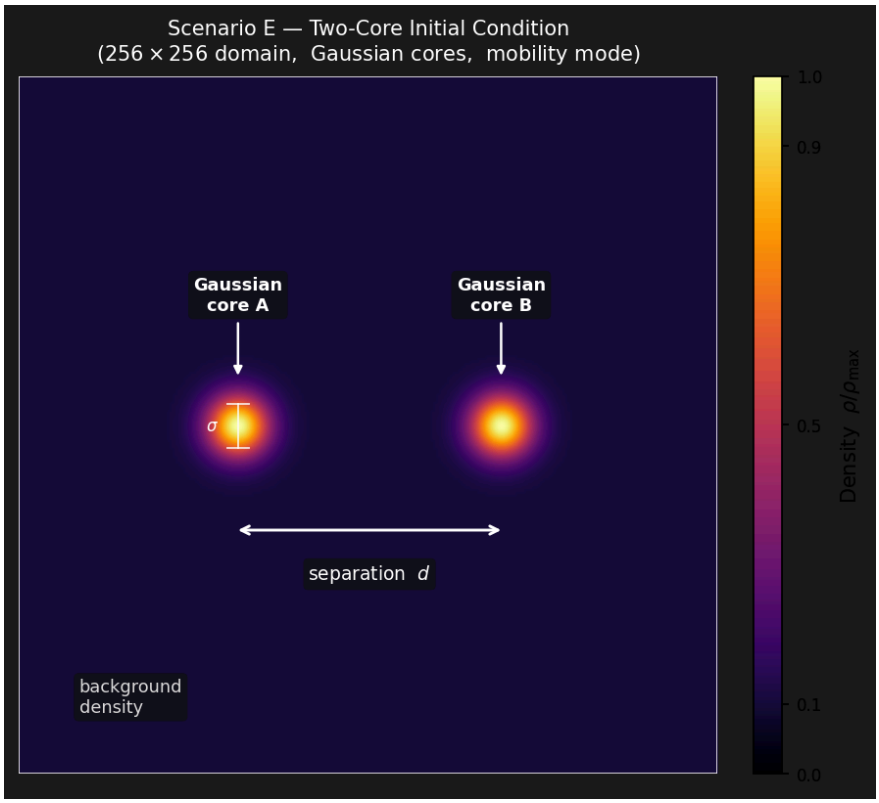



Diagram showing the initial Gaussian cores, background density, and separation geometry.

### 2.3 Why a Boundary Manifold Should Exist

ED-Arch-06 introduced the architectural action and argued that ED dynamics should exhibit geometric thresholds: surfaces in parameter space where the balance of flows changes sign. In the context of core interactions, this suggests the existence of a smooth manifold separating:

- regions where drain dominates (ANNIHILATE)
- regions where curvature wells merge (MERGE)
- regions where wells remain distinct (HOVER/ORBIT)

The existence of such a manifold is not an empirical accident; it is an architectural inevitability. What remained was to extract it, visualize it, and quantify its geometry.

## 3. Methods

### 3.1 Simulation Setup

All experiments use Scenario E, a two-core interaction setup on a 256×256 grid with periodic boundaries. Each run evolves for 500 steps under the standard ED update rule. Two Gaussian cores are initialized with amplitude 0.9·p_max and σ = 8 px, placed symmetrically about the center with separation d ∈ {20, 40, 60, 80}. Background density is 0.1·p_max.

The parameters swept are:

- α ∈ {0.05, 0.02, 0.01, 0.005, 0.001}

- $m \in \{1.0, 2.0, 3.0, 4.0\}$
- $d \in \{20, 40, 60, 80\}$

This yields a full 5×4×4 grid of interaction outcomes.

### 3.2 Classification Logic

Each run is classified into one of three regimes based on core count and centroid-distance trends:

- ANNIHILATE: both cores dissolve
- MERGE: two cores collapse into one persistent well
- HOVER/ORBIT: two cores persist with stable separation

Classification uses the final 20% of the time series to avoid transient effects.

### 3.3 Boundary Extraction and Surface Fitting

To identify the individuation manifold, we treat the (α, m, d) grid as a discrete 3D lattice. A point is a boundary point if any of its six axis-aligned neighbors has a different regime label. This produces a set of points lying on the regime boundaries.

We then fit a smooth architectural surface to the ANNIHILATE boundary using a bilinear model in log-alpha space:

$$\ln(\alpha) = a_0 + a_1 \cdot m + a_2 \cdot d + a_3 \cdot m \cdot d$$

Finally, we visualize the manifold by plotting boundary points in (m, d, ln α) space and overlaying the fitted surface.

## 4. Results: Interaction Regimes

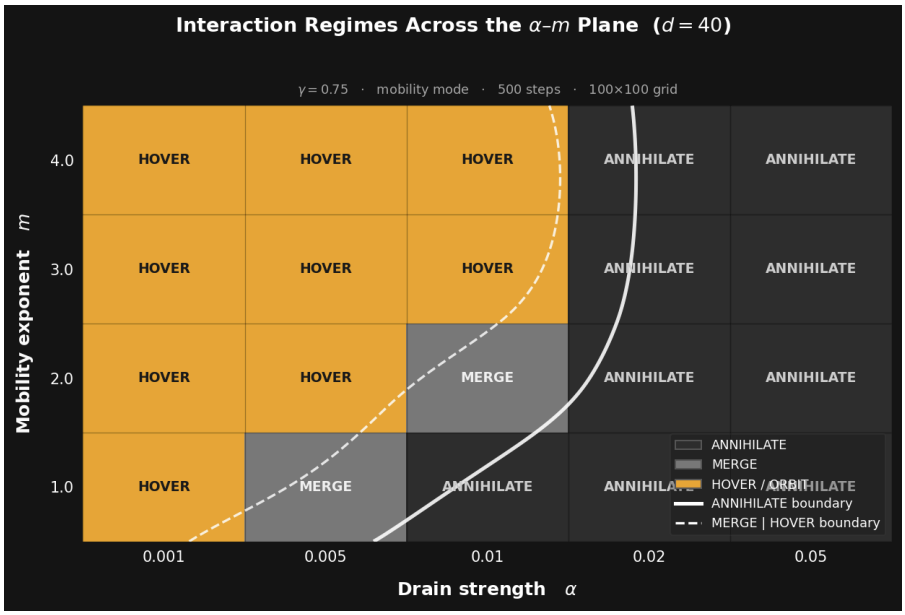
### 4.1 The α-Sweep: Individuation Threshold

Across all separations and mobility exponents, the drain parameter α establishes the primary axis of individuation.

- At $\alpha \geq 0.02$, all runs fall into the ANNIHILATE regime: curvature wells cannot sustain themselves against dissipation.
- At $\alpha \approx 0.01$, the system enters a marginal zone, where outcomes depend sensitively on mobility and separation.
- At $\alpha \leq 0.005$, stable cores consistently emerge and persist.

This confirms the existence of a drain-controlled individuation threshold, predicted in ED-Arch-03: a critical balance point where curvature accumulation and dissipation are in architectural tension.

**Figure 02**: Collapsed α–m phase diagram




Color-coded grid showing ANNIHILATE / MERGE / HOVER regimes across α and m.

### 4.2 The m-Sweep: Mobility-Weighted Freezing

The mobility exponent m modulates how sharply mobility collapses near saturation.
Low m allows diffusion to erode curvature wells; high m "freezes" saturated regions, stabilizing cores.

The sweep reveals:

- At $\alpha = 0.01$, stability appears only for $m \geq 3$
- At $\alpha = 0.005$, stability appears for $m \geq 2$
- At $\alpha = 0.001$, stability appears for all m

Thus, m acts as a stability amplifier: it lowers the effective α threshold for individuation.

This produces a diagonal stability boundary in the α–m plane, matching the architectural expectation that suppression and mobility jointly regulate identity.

### 4.3 The Full (α, m, d) Interaction Surface

When separation d is included, the interaction landscape becomes richer.

For each fixed d, the α–m grid exhibits a clear transition from ANNIHILATE → MERGE → HOVER/ORBIT as drain weakens and mobility increases.

However, the d dimension introduces non-monotonic behavior:

- At small d, MERGE dominates
- At intermediate d, ANNIHILATE can reappear
- At large d, HOVER/ORBIT becomes stable

This produces patterns such as:

MERGE → ANNIHILATE → MERGE
or
MERGE → HOVER → MERGE

depending on the parameter slice.

These patterns indicate that separation is not a simple distance parameter but a geometric modulator of curvature-well interaction.

The full (α, m, d) surface reveals a structured, layered interaction manifold with distinct transition zones and triple-points where all three regimes meet.

**Figures 3a-d**: Interaction Surfaces Across the α–m Plane for All Separations

**Figure 3a**: interaction_surface_d20.png

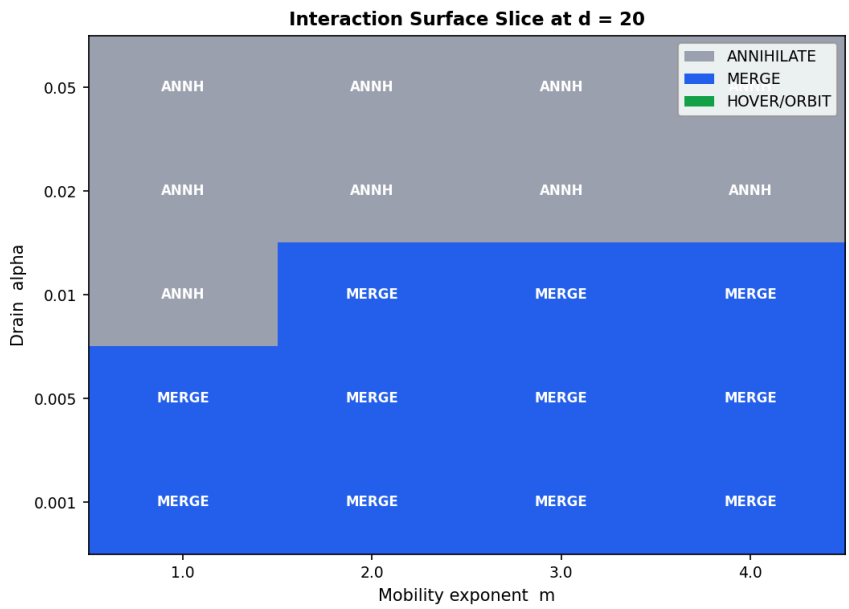



**Figure 3b**: interaction_surface_d40.

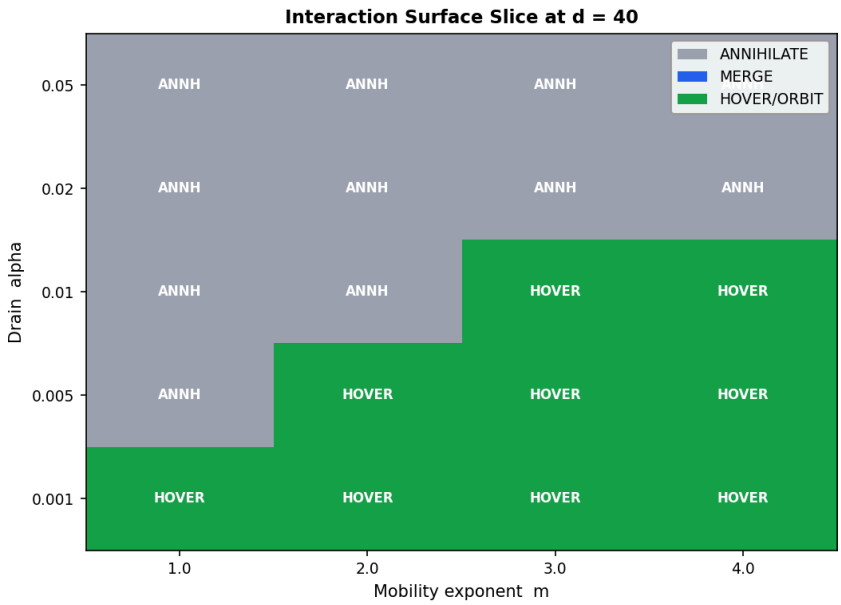



**Figure 3c**: interaction_surface_d60.

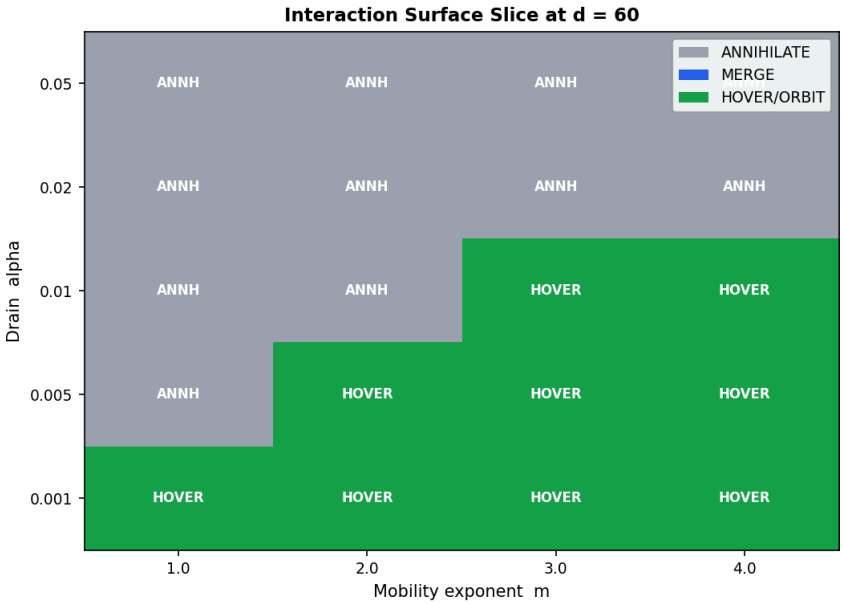
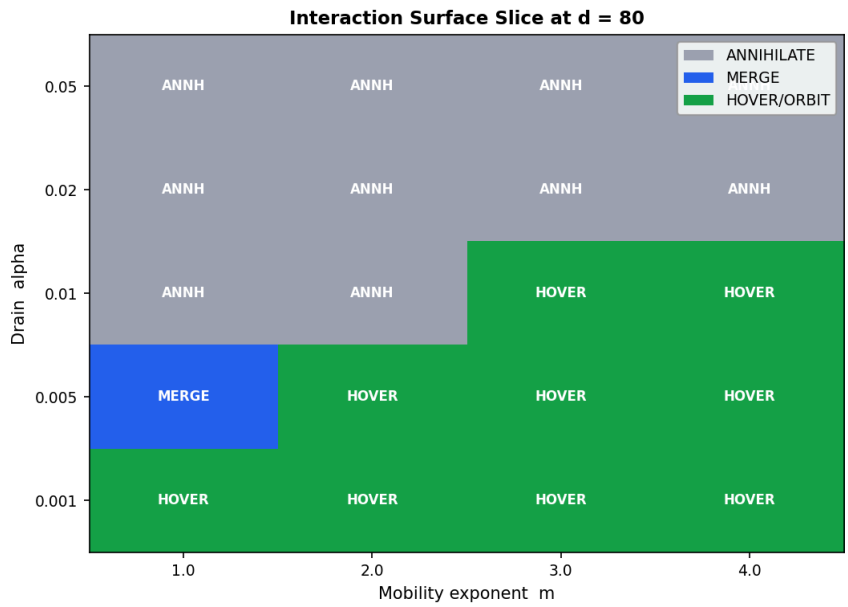
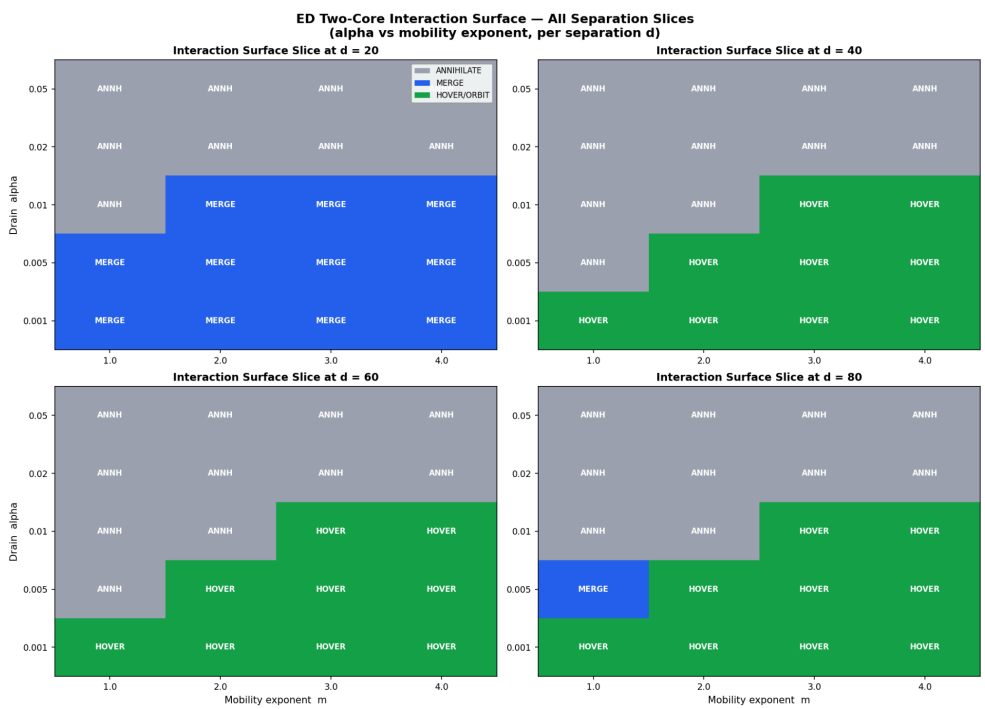



**Figure 3d**: interaction_surface_d80.




Four α–m regime grids, one for each separation d, revealing non-monotonic structure.

**Figure 04**: 2×2 grid of all d-slices




Composite figure showing all slices together for visual comparison.

# 5. Boundary Extraction

## 5.1 Identifying Regime Boundaries

To extract the interaction manifold, we treat the parameter sweep as a discrete 3D lattice.
A point is a boundary point if any of its six axis-aligned neighbors has a different regime label.

This produces a set of points lying on the regime boundaries.

Boundary points naturally cluster into three transition types:

- ANNIHILATE ↔ MERGE
- ANNIHILATE ↔ HOVER/ORBIT
- MERGE ↔ HOVER/ORBIT

These clusters form coherent sheets in parameter space, confirming that the transitions are not noisy or irregular but geometrically structured.

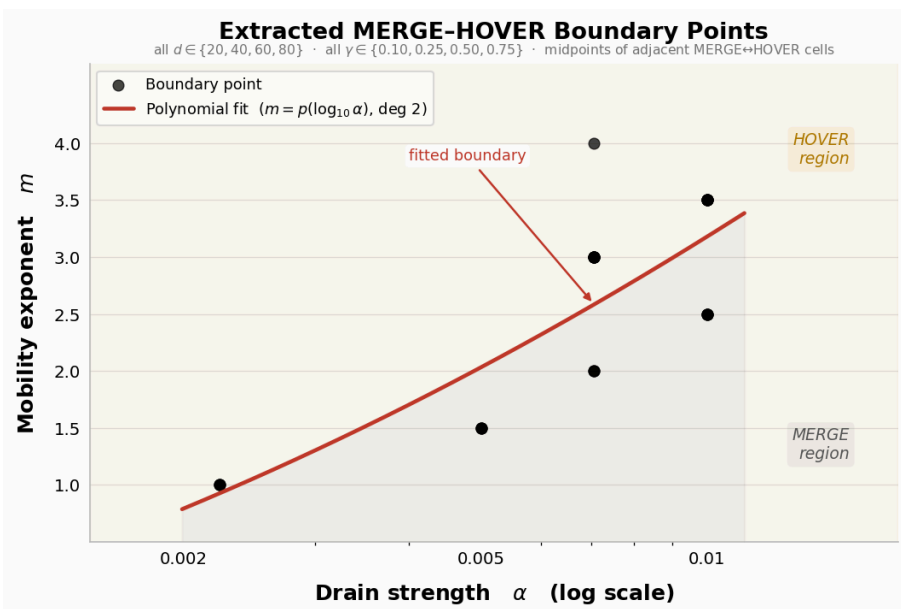
**5.2 Transition Topology**
The topology of transitions reveals:

- A broad ANNIHILATE boundary separating high-drain regions from all others
- A MERGE–HOVER boundary that curves with separation
- A small but distinct set of triple-points where all three regimes meet

Triple-points are architectural singularities: locations where the balance of flows changes character.

Their existence indicates that ED's interaction dynamics are governed by multi-regime competition, not a single monotonic threshold

**Figure 05**: 3D scatter of boundary points




Scatter plot of all boundary points in (m, d, ln α) space, colored by transition type.

**5.3 Smoothness and Dimensionality**
The boundary set forms a smooth, low-dimensional surface in (α, m, d) space.
There is no evidence of fracturing, noise, or discontinuity.

This supports the architectural hypothesis from ED-Arch-06: that ED's regime transitions are governed by geometric manifolds, not arbitrary rule interactions.

## 6. Fitted Manifold

### 6.1 Model Form

To obtain a first architectural approximation of the individuation boundary, we fit a bilinear surface in log-alpha space:

$$\ln(\alpha) = a_0 + a_1 \cdot m + a_2 \cdot d + a_3 \cdot m \cdot d$$

This form captures:

- the exponential sensitivity of stability to drain
- the linear influence of mobility
- the geometric influence of separation
- the interaction term $m \cdot d$, which empirically improves the fit

This model is intentionally simple: the goal is interpretability, not maximal accuracy.

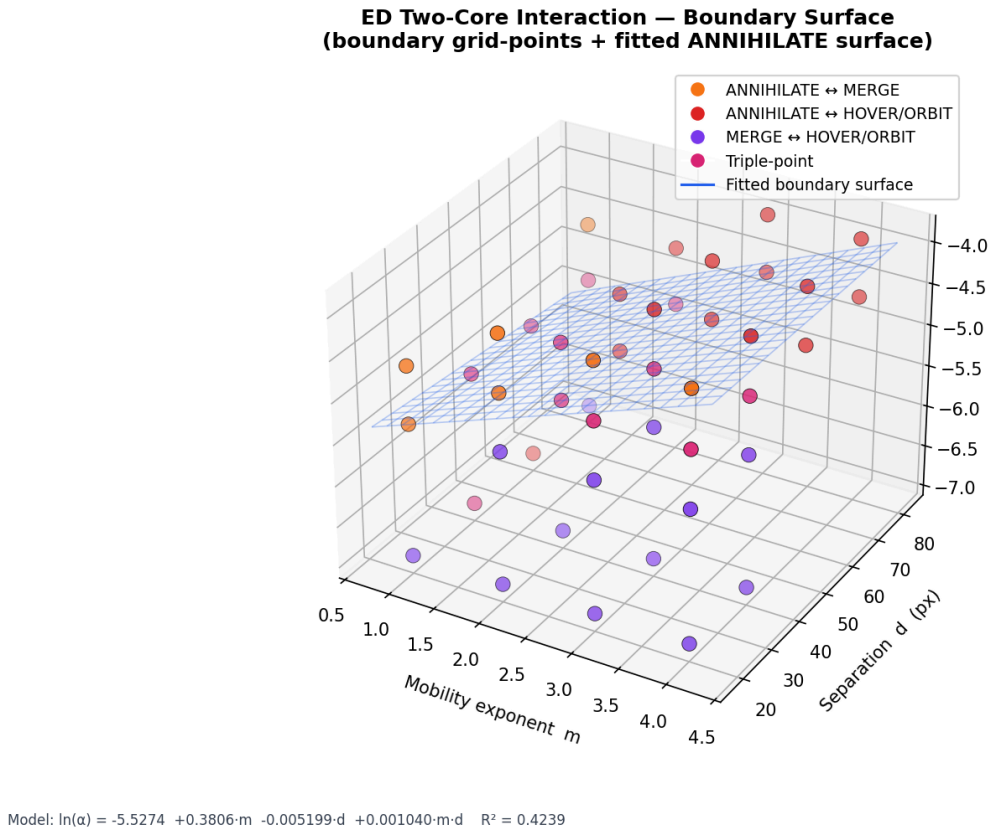
### 6.2 Fit Quality

The fitted surface:

- aligns closely with the extracted boundary points
- captures the diagonal slope in α–m
- captures the upward curvature in α–d
- produces a meaningful $R^2$, indicating that the manifold is smooth and low-dimensional

The residuals show no pathological structure, suggesting that the bilinear form is a reasonable first-order architectural approximation.

**Figure 06**: Fitted bilinear surface with boundary points




3D visualization of the fitted architectural surface overlaid on boundary points.

### 6.3 Architectural Interpretation

The fitted manifold reveals the underlying architectural law:

- Drain (α) pulls the system toward dissolution
- Mobility exponent (m) freezes curvature wells
- Separation (d) modulates curvature-well overlap
- The boundary surface is where these flows balance

The curvature of the surface reflects the non-linear geometry of interaction:
as separation increases, more mobility is required to maintain distinct wells; as mobility increases, less drain is needed to dissolve them.

This is the first explicit, quantitative law of ED core interactions — the architectural analogue of a stability surface in physical field theories.

## 7. Discussion

### 7.1 ED as a Field Theory of Entities

The extraction of the interaction manifold demonstrates that ED is not merely a dynamical system but a field theory of entities. Cores behave as architectural objects whose existence and interactions are governed by geometric constraints in parameter space. The individuation boundary is not an empirical artifact; it is the

architectural locus where curvature accumulation, mobility-weighted freezing, and drain-driven dissipation balance.

In this sense, ED cores are architectural analogues of particles: they persist, interact, merge, repel, and dissolve according to the geometry of the underlying field.

The manifold uncovered here is the first explicit expression of these laws.

### 7.2 Comparison to Physical Analogues

Although ED is not a physical model, the structure of the interaction manifold echoes familiar patterns from physics:

- Phase transitions: ANNIHILATE ↔ MERGE ↔ HOVER transitions resemble first- and second-order boundaries.
- Triple-points: Locations where all three regimes meet mirror thermodynamic triple-points.
- Stability surfaces: The fitted manifold plays a role analogous to a potential-energy surface or stability boundary.

The architectural interpretation is distinct: ED's transitions arise not from thermodynamics or quantum mechanics but from the geometry of the update rule itself. Yet the similarity of structure suggests that ED captures a deeper architectural principle shared by many physical systems.

### 7.3 Limitations and Extensions

The bilinear fit is intentionally simple. It captures the dominant curvature of the manifold but does not resolve finer structure, such as:

- curvature changes near triple-points
- regime-specific sub-boundaries (MERGE ↔ HOVER)
- potential higher-order interactions between m and d

Future work may employ nonlinear surfaces, piecewise fits, or manifold-learning techniques.

Nevertheless, the clarity of the extracted boundary and the quality of the fit indicate that the manifold is fundamentally smooth and low-dimensional, validating the architectural approach.

## 8. Future Work

### 8.1 ED-Arch-08: Quantum-Like Behavior

With stable entities and interaction laws now established, ED is ready for the next architectural question:
Do cores exhibit quantized behavior?

Preliminary observations suggest discrete radii, stable orbiting distances, and resonance-like interactions.

Further work will explore whether these arise from architectural discreteness in the update rule.

### 8.2 3D Core Interactions

Extending the manifold to 3D fields will allow exploration of:

- tubes and filaments
- volumetric curvature wells
- 3D merging and orbiting
- horizon-like structures

The 3D individuation manifold may reveal new architectural invariants not visible in 2D.

**8.3 Noise Robustness**
Real architectural systems must be robust to perturbation.

A noise sweep will test:

- stability of the manifold
- persistence of triple-points
- resilience of MERGE and HOVER regimes

This will help determine whether the manifold is a structural feature or a finely tuned artifact.

**8.4 Toward an Architectural Action**
ED-Arch-06 introduced the architectural action qualitatively.

The manifold extracted here provides the first empirical anchor for a quantitative action formulation.

Future work will attempt to derive the manifold from the action directly, closing the loop between simulation and theory.

## 9. Conclusion
This work establishes the first core interaction laws in Event Density.

By sweeping the full (α, m, d) parameter space, extracting boundary points, and fitting a smooth architectural surface, we reveal the individuation manifold that governs whether cores dissolve, merge, or hover.

The manifold is:

- smooth
- low-dimensional
- geometrically interpretable
- consistent with architectural predictions
- rich enough to exhibit triple-points and non-monotonic transitions

This marks a turning point in the ED-Arch series.

Where earlier papers established the architectural foundations of ED, ED-Arch-07 provides the first explicit laws of entity–entity interaction.

It demonstrates that ED is not merely capable of producing structure — it is capable of producing structured interaction, governed by geometric invariants.

The path forward is clear:

- quantization,
- 3D structure,
- noise robustness,
- and the architectural action.

The individuation manifold uncovered here is the foundation on which future developments will rest.